

Cancer as a Bounded Ecological Community

Cancer as a Bounded Ecological Community: A Program for Exposure-Aware, Ecologically Grounded, Twin-Steered Oncology

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Alternate Titles

1. Steering the Tumor Ecosystem: An Integrative Framework Linking Exposure, Ecology, and Adaptive Therapy
 2. The Three-Scale Architecture of Cancer: Exposure, Ecosystem, and Assimilation as a Unified Program
 3. From Carcinogen Flux to Adaptive Control: A Cancer Ecology Center Program for Individualized Oncology
 4. Beyond Eradication: How Reading Cancer at Three Coupled Scales Rewires the Therapeutic Mission
 5. The Bounded Tumor Town: Exposure-Aware, Ecologically Grounded, Computationally Steered Oncology
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Abstract

Cancer kills approximately ten million people per year and is projected to kill eighteen million annually by 2050 [IARC 2024; GBD Collaborators 2025]. Despite half a century of molecular oncology, the dominant clinical framework remains cohort-average: risk is assessed by family history and population-level hazard ratios, the tumor is treated as a growing object rather than a functioning ecosystem, and the patient's lifetime of carcinogenic exposures is absent from quantitative models. This white paper argues that these failures share a common root: cancer has not been read at the correct scales, with the correct models, in the correct combination.

The Cancer Ecology Center program advances a three-layer architecture that corrects this jointly. The **Exposure layer** (Exposograph/GeoToxGraph/IsoMCED) reads the carcinogenic flux each person has actually experienced across 14 IARC Group 1 carcinogen classes, 8 tissues, and individual genotype combinations that produce 11- to 33-fold variation in risk score between population percentiles and up to 47-fold variation across genotype combinations [Kazi, Sarigiannis, Pienta 2026]. The **Ecosystem layer** (Prairie-Dog/G-function) reads the tumor not as a diffusing invasion front but as a bounded, habitat-engineering, resource-depleting community whose realized expansion is 0.1–1.7% volume per day — one to two orders of magnitude below intrinsic potential — exactly as prairie dog towns expand at 30–40 meters of front per year far below their reproductive capacity [Pienta, Brown, Amend, Gatenby 2026]. The **Assimilation layer** (Cancer Ecology Digital Twin, CEDT) reads the patient continuously, ingesting imaging, circulating tumor DNA, tissue genomics, and laboratory panels into a state-space framework whose latent state vector $\mathbf{X(t)} = [\mathbf{H(t)}, \mathbf{E(t)}, \mathbf{T(t)}, \mathbf{M(t)}, \mathbf{Q(t)}]$ carries host, exposure, tumor-intrinsic, tumor-ecosystem, and state-quality subvectors and re-estimates the ecological interaction matrix $\mathbf{A(t)}$ at every clinical visit via Bayesian data assimilation [Pienta 2026].

The through-line at every scale is **steering, not eradication**: reroute carcinogen flux through detoxification networks; steer tumor ecology with adaptive therapy; continuously assimilate new data rather than predict from a fixed model. This program has specific, falsifiable, computational, and clinical deliverables. This white paper is the narrative architecture that unifies them.

Keywords: cancer ecology; exposome; carcinogenomics; ExposoGraph; prairie dog towns; G-function; adaptive therapy; digital twin; Bayesian data assimilation; generalized Lotka-Volterra; IsoMCED; GeoToxGraph; cancer ecology center; steering; bounded ecological community

Executive Summary

The Problem Is Structural

Approximately 10 million people died of cancer in 2022 [CEDT v6.5, ref 136]. The International Agency for Research on Cancer and the 2025 Lancet Global Burden of Disease analysis project 35 million new diagnoses and 18 million deaths per year by 2050 [Sung et al. 2021; GBD Collaborators 2025]. These numbers are not primarily a failure of drug discovery. They are a failure of the decision-making architecture around drugs: risk prediction tools that ignore most of what made cancer happen; tumor models that treat a functioning ecosystem as a homogeneous growing sphere; treatment algorithms that optimize for average populations, not individual trajectories; and computational twins that carry no memory of the patient's exposure history and no capacity to track the tumor as an evolving community.

The Argument

Cancer is a bounded, engineered ecological community. It arises from a combinatorial flux of carcinogenic exposures processed through a genotype-specific metabolic network. Once established, it behaves not as a continental invader but as a prairie-dog town: bounded, habitat-engineering, resource-depleting, edge-managed, internally regulated, and vulnerable to collapse through ecological intervention rather than military eradication. And it can be forecast, in an individual patient, by a data-assimilating digital twin whose state-space explicitly carries the exposure history and the tumor ecosystem as first-class variables, not afterthoughts.

This three-layer reading — **Exposure (Exposograph), Ecosystem (Prairie-Dog/G-function), Assimilation (CEDT)** — is not a metaphor program. Every connection is mathematical: the carcinogen-flux burden vector E_{path} computed by ExposoGraph is literally the $E(t)$ subvector of CEDT's latent state. The behavioral dynamics of the tumor ecosystem — bounded growth, resource depletion, edge-interior phenotype partition — are the biological rationale for the generalized Lotka-Volterra (GLV) core, the resource compartments, and the

carrying-capacity structure of CEDT. The G-function of Vincent and Brown formalizes adaptive therapy as a control problem that CEDT individualizes for each patient. The IsoMCED isotope assay provides a non-invasive longitudinal observation channel for the H+E substate at population scale, deployable globally at under \$20 per sample. These are not parallel tracks; they are a coupled system.

The Through-Line: Steering, Not Eradication

The single most consequential reframe in this program is the shift from eradication to steering. You do not eradicate carcinogen exposure; you reroute flux through detoxification networks. You do not eradicate the tumor; you steer its ecology, maintaining a sensitive-cell reservoir that ecologically suppresses resistance. You do not predict cancer with a fixed model; you continuously assimilate new observations and update the forecast. Steering is not a softer goal than eradication; it is a more sophisticated one. It demands quantitative models, longitudinal data, and an individualized decision architecture that population-average oncology structurally cannot provide.

What the Cancer Ecology Center Is Uniquely Positioned to Deliver

The Cancer Ecology Center at Johns Hopkins holds the only research program that simultaneously develops and integrates all three layers. Exposograph 2.0 is under review at PLOS Computational Biology. The CEDT manuscript is in final preparation for submission. The Prairie-Dog Perspective is in revision. GeoToxGraph is released as an open-source browser. IsoMCED has pilot funding through a Fast Grant and a commercial translation pathway through Krefect, Inc. The CarcinoGenomics R01 (Research Strategy v61) is in active NIH submission. A 16-institution Cancer Ecology Consortium operationalizes the program at scale.

In the next decade, the Cancer Ecology Center will: complete the Hopkins Biobank for Genomics and Exposomics (BGES, n = 1,000); validate Exposograph predictions against All of Us and NHANES; complete a prospective CEDT validation on an intermittent-therapy prostate cancer cohort; extend adaptive therapy to a second tumor type; and deliver IsoMCED through regulatory qualification toward a globally deployable cancer screening instrument. Every component will be released as open science under MIT/CC-BY licensing.

The field has built the components. The Cancer Ecology Center integrates them into a program.

Section 1. The Problem: Why Cohort-Average Oncology Cannot Steer Disease in an Individual

1.1 The Burden and the Trajectory

In 2022, approximately 20 million people were newly diagnosed with cancer worldwide, and approximately 10 million died of the disease — roughly 27,000 deaths every day [CEDT v6.5, ref 136]. The International Agency for Research on Cancer and a 2025 Lancet Global Burden of Disease analysis project 35 million new cases and 18 million deaths annually by 2050 [Sung et al. 2021; GBD Collaborators 2025]. These numbers are not predictions about drug efficacy; they are projections about incidence and coverage, anchored in aging demographics, global pollution burden, and the continuing expansion of industrial chemical exposure into populations with no historical adaptation to those chemicals.

The oncology enterprise has responded to this trajectory with enormous investment in targeted therapies, immunotherapies, and precision genomics. These investments have produced real progress in specific tumors and specific patient populations. They have not bent the aggregate mortality curve. The reason is structural: the dominant clinical and computational decision architecture was built for average populations, not individual trajectories. It is excellent at telling a clinician what has been true for patients like this one in the past. It is structurally incapable of telling a clinician what will happen next in this patient under each of several candidate treatment paths.

Three blind spots define this structural incapacity.

1.2 Blind Spot 1: The Exposome Is Collapsed to a Checkbox

Every oncology staging system, every tumor board workflow, and every prognostic model in routine clinical use asks about tobacco (pack-years), alcohol (drinks per week), and occupational exposure (binary yes/no for a handful of industries). None of them capture what is actually happening at the level of carcinogen metabolism: which IARC Group 1 carcinogens the patient has been exposed to across a lifetime, in what doses, in what combinations, processed through which genotype-specific enzyme variants, in which tissues.

The result is that the most important causal information about why this cancer arose in this person is absent from every model used to forecast what will happen next. ExposoGraph 2.0 demonstrates that this omission is quantitatively consequential: modeling 10,000 synthetic genomes across five ancestry groups, it finds 11- to 33-fold variation in cancer risk score between population percentiles and up to 47-fold variation across genotype combinations, driven by variation in CYP, NAT, GST, ALDH, SULT, and UGT enzyme variants, and by the nonlinear interaction of concurrent multi-carcinogen exposures [Kazi, Sarigiannis, Pienta 2026]. Single-carcinogen, single-gene, population-average models cannot capture this range because they are not architecturally designed to.

1.3 Blind Spot 2: The Tumor Microenvironment Is Treated as a Fixed Parameter

Most oncology decision tools — and most oncology digital twins — represent the tumor as a single growing object. Drug sensitivity is a fixed parameter or an efficacy scalar derived from population-average trial data. The immune compartment, the stromal and vascular microenvironment, the resource landscape, and the subclonal competition that drives resistance are either absent or collapsed to a single summary statistic.

This architecture is wrong about what tumors are. A tumor is an ecosystem: a mosaic of competing subclones preyed upon by anti-tumor immune populations, facilitated by pro-tumor immune and stromal populations, dependent on vasculature that gates drug delivery, and shaped by resource availability that determines which phenotypic strategies survive [Dujon et al. 2021; Aktipis et al. 2015; Lloyd et al. 2016]. The interaction matrix among these populations — who suppresses whom, who facilitates whom, at what rates — is not fixed. It evolves during treatment, shifts across the tumor's edge-interior axis, and resets after perturbation. A model that treats these dynamics as fixed parameters cannot forecast trajectory or individualize treatment decisions.

1.4 Blind Spot 3: The Model Is Static Rather Than Data-Assimilating

Even the best mechanistic digital twins published in oncology are calibrated once, at baseline, and then run forward. They produce a single forecast trajectory from a fixed initial condition. This architecture is deeply mismatched to the clinical reality of cancer: a

patient typically receives multiple observation modalities — imaging, circulating tumor DNA, tissue genomics, laboratory panels — at irregular, sparse intervals over months to years. Each observation updates what is known about the disease state. A model that does not continuously revise its estimate of the latent state — the interaction matrix, the immune phenotype, the resource landscape — wastes most of the information the clinical workflow already generates.

The right architecture is data assimilation: a framework that maintains a posterior distribution over the latent state, propagates it forward with a dynamical model between observations, and updates it with each new observation using a Bayesian filter. This is the architecture of CEDT [Pienta 2026]. It is also the architecture used for decades in weather forecasting, financial risk modeling, and aerospace tracking — domains that accept that forecasting under uncertainty requires continuous updating, not a single calibration at time zero.

1.5 The Cost of the Three Blind Spots Together

Population-average risk assessment misses the individuals at highest genetic and exposure risk. Population-average tumor modeling misses the ecological dynamics that determine resistance. Static calibration wastes the longitudinal information that clinical workflows already collect. The three blind spots compound: without individualized exposure history, the prior for the initial tumor state is wrong; without ecosystem modeling, the forward projection is wrong; without data assimilation, the errors accumulate rather than correct. The result is an oncology decision architecture that is systematically wrong for the individual patient in ways that cannot be corrected by adding more drugs to the formulary.

The Cancer Ecology Center program is built to correct all three blind spots simultaneously, in a coupled, computable, and falsifiable architecture.

Section 2. A Brief History of the Idea

2.1 The Ecological Reframe Has Deep Roots

The recognition that cancer is not simply a genetic accident but an ecological and evolutionary process has accumulated over seven decades, through a series of partial but important insights that the Cancer Ecology Center program now integrates.

Julian Huxley proposed in 1956 that a malignant tumor, once it crosses the threshold of cellular autonomy, can be regarded as a new biological species [Huxley 1956]. The analogy was prescient: a cancer cell is not merely a broken normal cell; it is a lineage-distinct organism, reproductively isolated from the host's differentiated cells, competing for resources, and selected on its own fitness. Huxley's framing predates the molecular revolution but anticipates its evolutionary synthesis.

Peter Nowell formalized the mechanistic basis for this view in 1976, describing the origin of tumor cell heterogeneity as a product of clonal evolution under natural selection [Nowell 1976]. Random mutations generate heritable variation; the microenvironment selects on that variation; the surviving clone diversifies further as the environment shifts. Clonal evolution is now understood to produce the intratumoral heterogeneity that drives treatment resistance: not all cells are equally drug-sensitive, and selection pressure from therapy accelerates the rise of resistant subclones.

Tomlinson and Bodmer extended this framework in 1995, demonstrating that somatic selection among cells with varying fitness consequences is sufficient to explain tumor progression without invoking rare sequential mutations in specific genes [Tomlinson & Bodmer 1995]. John Cairns had made a related argument in 1975, proposing that the architecture of normal epithelia — organized with asymmetrically dividing stem cells whose daughters are rapidly shed — is an evolutionary adaptation against somatic mutation accumulation [Cairns 1975]. Cancer, in this view, is a breakdown of an evolved anti-cancer cellular governance system.

2.2 Game Theory, the G-Function, and Adaptive Therapy

The transition from evolutionary description to quantitative prediction required mathematical formalism. Thomas Vincent and Joel Brown provided this in 2005 with the G-function framework: a fitness-

generating function $G(v; u, x, E)$ that separates the heritable strategy v of an individual from the strategies u of its competitors, their abundances x , and the environmental state E [Vincent & Brown 2005]. The G -function is not merely a metaphor apparatus; it generates two coupled differential equations — one ecological (governing population sizes) and one evolutionary (governing strategy values) — and places adaptive therapy on rigorous game-theoretic ground.

Robert Gatenby and colleagues translated this formalism into clinical protocol in 2009, proposing that maintaining sensitive-cell populations in competition with resistant subclones — rather than driving for maximum cytoreduction — would delay resistance [Gatenby et al. 2009]. Jingsong Zhang and colleagues tested this in a prospective clinical trial in metastatic castration-sensitive prostate cancer, demonstrating in 2017 that adaptive dosing of abiraterone, triggered by PSA threshold response, extended time to progression compared to historical continuous-dosing controls [Zhang et al. 2017]. This is the first published prospective validation of evolutionary steering in oncology.

2.3 The Ecosystem Turn

Athena Aktipis and colleagues provided the conceptual synthesis of tumor biology and multicellularity theory in 2015, arguing that cancer is best understood as a breakdown of the five hallmarks of multicellular cooperation: controlled proliferation, controlled cell death, resource sharing, division of labor, and environmental maintenance [Aktipis et al. 2015]. This framing locates cancer not as an aberration of the individual genome but as a reversion to a more ancient, unicellular fitness landscape — the ecosystem beneath the organism.

Antoine Dujon, Robert Gatenby, Beata Ujvari, and colleagues synthesized the ecological evidence in 2021, arguing explicitly that tumors are ecosystems and that cancer ecology — the study of interactions among tumor cells, immune cells, stroma, and host — should be treated as a scientific discipline in its own right [Dujon et al. 2021]. This synthesis anchors the tumor-as-ecosystem reading that this program takes as its starting point.

2.4 The Exposome and Its Measurement Problem

Christopher Wild coined the term “exposome” in 2005 to describe the totality of environmental exposures from conception onward, arguing that genomics without exposomics provides an incomplete picture of disease etiology [Wild 2005]. A 2025 retrospective at the twentieth anniversary of the exposome concept confirmed that the measurement problem — how to capture a lifetime of diverse exposures in a single deployable instrument — remains the central unsolved challenge [Wild et al. 2025].

The IsoMCED instrument developed in the Cancer Ecology Center program addresses this measurement problem directly: stable isotope ratios (δ -¹³C, δ -¹⁵N, δ -³⁴S) in hair keratin provide a non-invasive, globally deployable record of cumulative carbon, nitrogen, and sulfur metabolism that integrates exposure, host metabolism, and tumor metabolism in a single sample [Pienta Fast Grant 2026]. Hair has no cold chain requirements, requires no phlebotomy, and can be processed for under \$20 per sample at isotope ratio mass spectrometry facilities — making it feasible for low- and middle-income country deployment at population scale.

2.5 Clonal Architecture and Spatial Heterogeneity

Marco Maley, Mel Greaves, Kornelia Polyak, and colleagues developed the clonal architecture of solid tumors through the 2000s and 2010s, demonstrating that most advanced cancers harbor dozens to hundreds of coexisting subclones distributed heterogeneously in space [Maley et al. 2017; Greaves & Maley 2012]. Mark Lloyd and colleagues demonstrated in 2014 and 2016 that the spatial distribution of phenotypes within an invasive ductal breast carcinoma recapitulates a food-safety tradeoff: edge cells maximize motility and invasion at the cost of proliferative rate, while interior cells maximize proliferation in the resource-depleted hypoxic core [Lloyd et al. 2014; Lloyd et al. 2016]. This edge-interior phenotype partition is one of the structural features shared with prairie dog towns and formalized in the CEDT ecosystem layer.

2.6 Pienta and the Speciation Framing

Kenneth Pienta and colleagues argued in 2020 that cancer constitutes speciation in a biologically meaningful sense: the cancer cell becomes a lineage-distinct organism that is, under the rules of evolutionary biology, a new species in the environment of the patient [Pienta et

al. 2020]. This framing is not merely terminological; it implies that the tools of ecology — niche theory, competition and coexistence, resource modeling, game theory — are the correct analytical framework for understanding cancer behavior, and that therapeutic interventions are best understood as ecological perturbations rather than chemical warfare.

Each of these intellectual strands — evolutionary speciation, clonal evolution, game-theoretic strategy, ecosystem engineering, exposome measurement, adaptive therapy — was necessary for the current program. None alone is sufficient. The Cancer Ecology Center program integrates them into a coupled, computable architecture that is more than the sum of its parts.

Section 3. Origin: The Exposograph Layer

3.1 Cancer Risk Is a Combinatorial Flux Problem

Cancer does not arise from a single carcinogen acting on a single susceptible gene in an otherwise uniform cellular population. It arises from a combinatorial flux of environmental exposures, processed through genotype-specific enzyme networks, in tissue-specific expression contexts, over time scales of years to decades. The conventional epidemiological framing — assigning population-attributable fractions to single agents, estimating relative risk from ever/never exposure dichotomies — captures the aggregate signal but misses the individual variation that determines who actually develops cancer.

ExposoGraph 2.0 is the Cancer Ecology Center's answer to this framing problem. It is a quantitative, computable, tissue-specific platform for modeling multi-carcinogen interaction and individualized risk assessment across 14 IARC Group 1 carcinogen classes and eight cancer-relevant tissues [Kazi, Sarigiannis, Pienta 2026]. The platform integrates seven computational modules: a 156-node, 208-edge knowledge graph spanning 14 carcinogen classes; tissue-specific graph views derived from GTEx v8 expression data; a Michaelis-Menten flux engine with genotype-dependent modifiers; three-tier exposure integration using NHANES, EPA IRIS, and OSHA surveillance data; a multi-carcinogen interaction engine modeling competitive inhibition,

glutathione depletion, and induction cascades; a population-scale simulator with an All of Us adapter; and a clonal hematopoiesis of indeterminate potential (CHIP) somatic modifier layer.

3.2 Quantitative Results from ExposoGraph 2.0

The results from ExposoGraph 2.0 are not hypothetical; they are computed from published databases and validated against published enzyme kinetic parameters. Tissue-specific graph construction from GTEx v8 expression data reveals that the degree of pathway preservation varies substantially across tissues: hepatic carcinogen-metabolism graphs retain 94% of edges present in the full network, while mammary tissue graphs retain only 67% [Kazi, Sarigiannis, Pienta 2026]. This means that a carcinogen-metabolizing pathway present in the liver may be entirely absent in breast tissue — a fact with direct implications for tissue-specific risk and for the tissue specificity of carcinogen-induced cancers.

The kinetic engine calculates metabolic flux ratios spanning three orders of magnitude based on genotype combinations across CYP, NAT, GST, ALDH, SULT, and UGT enzyme families [Kazi, Sarigiannis, Pienta 2026]. The multi-carcinogen interaction engine consistently produces super-additive risk from concurrent exposures: synergy scores range from 1.2 to 2.38 across modeled exposure pairs, with a glutathione depletion threshold beyond which detoxification fails nonlinearly — meaning that exposures that are individually manageable become collectively catastrophic above a threshold GSH concentration [Kazi, Sarigiannis, Pienta 2026].

Population simulations of 10,000 synthetic genomes across five gnomAD v4.1 genetic ancestry groups produce log-normal risk score distributions with 11- to 33-fold variation between the 5th and 95th percentiles and up to 47-fold variation across all genotype combinations [Kazi, Sarigiannis, Pienta 2026]. These ancestry-specific risk architectures are driven by population allele frequency profiles that differ substantially across gnomAD ancestry groups for the relevant pharmacogenomic variants. A single-carcinogen, single-gene model cannot capture this range; it is structurally incapable of doing so because it does not model either the combinatorial interaction of exposures or the genotypic variation in processing enzymes.

3.3 Tissue Network Topology and the CHIP Layer

The tissue-specificity of carcinogen-mediated risk is not merely a question of exposure route or dose. It is a question of network topology: which carcinogen-metabolism enzymes are expressed in which tissues, in what combinations, and at what levels. ExposoGraph 2.0 computes tissue-specific graphs for eight cancer-relevant tissues from GTEx v8 data, enabling direct comparison of how the same carcinogen is processed in the liver versus the lung, the bladder versus the breast. The structural consequences for risk are substantial and largely unaccounted for in current population-level risk models.

CHIP — clonal hematopoiesis of indeterminate potential, the age-associated accumulation of somatic mutations in hematopoietic stem cells — is modeled in ExposoGraph 2.0 as a somatic modifier layer that alters immune-cell carcinogen processing capacity over time [Kazi, Sarigiannis, Pienta 2026]. CHIP variants in DNMT3A, TET2, and ASXL1 alter myeloid cell inflammatory phenotype and oxidative stress handling in ways that interact with carcinogen-induced DNA damage signaling. This layer connects the carcinogenomics of cancer origin to the immune-microenvironment dynamics of cancer behavior — a bridge between the Exposograph and Prairie-Dog layers.

Figure 1 — [PLACEHOLDER] ExposoGraph 2.0 Multi-Layer Architecture. A three-panel figure illustrating: (A) The 156-node, 208-edge knowledge graph with 14 carcinogen classes connected through bioactivation, detoxification, and DNA-damage nodes; (B) Tissue-specific graph views comparing hepatic (94% edge retention) and mammary (67% edge retention) networks, with the pathway-absent nodes highlighted; (C) Population-scale risk score distribution across five gnomAD ancestry groups, showing log-normal distributions with 11- to 47-fold variation in predicted cancer risk. Caption: ExposoGraph 2.0 transforms the exposome from a checkbox to a quantitative, tissue-specific, genotype-dependent risk landscape. The 47-fold range in risk scores across individuals is the quantitative motivation for personalized exposure assessment [Kazi, Sarigiannis, Pienta 2026].

3.4 The GeoToxGraph Contrast: Why Humans Are Uniquely Vulnerable

Understanding why specific chemicals are carcinogenic to humans — and not to the environmental microorganisms that also encounter them — requires an evolutionary contrast framework. The GeoToxGraph provides this framework through evolutionary contrast

mapping: a graph-based approach to comparing microbial and human exposure metabolism at the level of shared chemistry, divergent biological implementation, and pathway presence versus absence [Pienta Geobacter Lab 2026].

Geobacter and related environmental microorganisms encounter many of the same chemicals as human cells: arsenic and chromium in groundwater, polycyclic aromatic hydrocarbons in combustion residues, chlorinated solvents in contaminated aquifers, cadmium and mercury in industrial outflows [Pienta Geobacter Lab 2026]. The chemistry is shared; the biology that engages it is not. Geobacter strains can use metals and organohalides as terminal electron acceptors — Cr(VI) and U(VI) as respiratory substrates, chlorinated ethenes as electron acceptors, aromatic substrates through the benzoyl-CoA pathway toward anaerobic dearomatization. Human cells cannot perform these microbial respiratory or anaerobic-catabolic routes. They rely instead on cytochrome P450 oxidation, glutathione conjugation, methylation, metallothionein induction, ABC-transporter efflux, DNA damage response, and tissue-level injury.

The GeoToxGraph formalizes this divergence through a six-class contrast ontology: conserved chemistry, analogous function, pathway loss, toxic inversion, host-shifted handling, and ecosystem-outsourced [Pienta Geobacter Lab 2026]. The critical classes for understanding human carcinogenesis are **pathway loss** and **toxic inversion**. Pathway loss occurs where microbes possess a catabolic or respiratory route that humans lack entirely — as with organohalide respiration (Geobacter can respire on TCE/PCE as electron acceptors; human cells cannot dechlorinate them, leaving the CYP/GSH bioactivation route to produce reactive epoxide intermediates). Toxic inversion occurs where the same chemical undergoes opposite biological outcomes in microbes versus humans — as with Cr(VI), which Geobacter reduces extracellularly to immobilized Cr(III) as a detoxification/energy-generating reaction, while human cells reduce Cr(VI) intracellularly to Cr(III) in a process that generates DNA-damaging reactive oxygen species and Cr-DNA adducts.

The current GeoToxGraph contrast layer contains 60 nodes and 82 edges; the strain-resolved layer contains 114 nodes and 147 edges, spanning a 22-compound panel of metals/metalloids, aromatic hydrocarbons, organohalides, and classical human-only carcinogen-metabolism anchors [Pienta Geobacter Lab 2026]. The framework is

released as an open-source browser at kpienta-cloud.github.io/geobacter/ and as CSV/GraphML data at github.com/kpienta-cloud/geobacter.

The interpretive payoff of the GeoToxGraph is this: humans are not simply less efficient at detoxifying certain chemicals. In the cases of pathway loss and toxic inversion, they use the chemistry of those chemicals in a direction that is actively harmful — producing DNA-damaging intermediates from substrates that environmental microorganisms respire for energy. This is why “the toxic landscape humans face” is itself an evolutionary contrast map; it encodes the history of pathway loss since the divergence from microbial ancestors.

3.5 IsoMCED: The Integrating Instrument

The ExposoGraph computes what an individual’s exposure and genotype imply about their carcinogen flux. The GeoToxGraph situates that flux in an evolutionary framework that explains why specific chemical classes are dangerous to human cells. What remains is an instrument that reads the actual biological output of these processes in a non-invasive, population-scalable format.

IsoMCED (Isotope Multi-Cancer Early Detection) is that instrument. It uses stable isotope ratios — delta-13C, delta-15N, and delta-34S — measured in hair keratin by isotope ratio mass spectrometry (IRMS) to read the cumulative output of carbon, nitrogen, and sulfur metabolism across the body [Pienta Fast Grant 2026]. Hair has several properties that make it uniquely useful for this application: keratin is metabolically stable once deposited, so it retains an isotopic record of the metabolic state at the time of growth; hair grows at approximately 1 centimeter per month, providing a temporal record without phlebotomy or biopsy; and hair requires no cold chain for collection, storage, or shipping, making it feasible for deployment in low- and middle-income country settings at under \$20 per sample including IRMS analysis.

Cancer reprograms host carbon, nitrogen, and sulfur metabolism in specific and measurable ways. Tumor cells preferentially consume ^{12}C -enriched glucose through aerobic glycolysis (the Warburg effect), depleting host tissues of ^{12}C and shifting bulk tissue and hair delta-13C. Elevated nitrogen catabolism from muscle wasting (cancer cachexia) shifts hair delta-15N upward. Sulfur metabolism shifts that accompany cancer-associated glutathione depletion and altered cysteine-methionine cycling appear in delta-34S. The Pienta lab generated the first BGES (Biobank for Genomics and Exposomics at

Hopkins) isotope-fractionation dataset linking these signatures to prostate cancer; Krefect, Inc. was spun out of this work to develop the first stable-isotope hair laboratory-developed test (LDT) in a CAP/CLIA setting [Pienta Fast Grant 2026].

The IsoMCED Fast Grant (Spring 2026) funds profiling of delta-13C, delta-15N, and delta-34S in 600 archived hair samples from six cancer types (prostate, breast, colorectal, lung, pancreatic, ovarian) plus controls from the Johns Hopkins biobank, training and benchmarking gradient-boosted and transformer classifiers, and pre-registration of an analytical validation protocol aligned to CE-IVDR Annex II and WHO PQDx evidence requirements [Pienta Fast Grant 2026].

3.6 Section Close: The Question Exposograph + GeoToxGraph + IsoMCED Answers

Together, these three instruments answer a question that single-chemical, single-gene, population-average models structurally cannot: what is the per-person, per-tissue, per-exposure-history risk landscape for this individual? Exposograph provides the mechanistic forward model — here is what this genotype and exposure history imply about carcinogen flux through detoxification networks. GeoToxGraph situates the most dangerous chemical classes in an evolutionary framework that explains why they are dangerous. IsoMCED reads the cumulative biological output of all of the above in a single non-invasive hair sample. The three instruments compose the Exposure layer of the unified program. Their output — the carcinogen-flux burden vector E_{path} — is not merely a risk estimate; it is a state variable that feeds directly into the CEDT as the $E(t)$ subvector of the latent state.

Section 4. Behavior: The Prairie-Dog Town Layer

4.1 The Quantitative Mismatch That Changes Everything

Tumor expansion is routinely cast in the language of invasion: tumors invade, metastases colonize, therapies eradicate. The Fisher-Skellam reaction-diffusion model gives the mathematics: a constant front velocity $v = 2\sqrt{rD}$, where r is the per-capita growth rate and D is the dispersal coefficient. Calibrated to invasive mammals and amphibians, this model produces front velocities in the right order of magnitude for

cane toads (over 50 km/year at the Australian front), muskrats (6-32 km/year in Europe), mink in Poland (approximately 15 km/year), grey squirrels in Britain and Ireland (approximately 5 km/year), and Australian rabbits (approximately 50 km/year) [Pienta, Brown, Amend, Gatenby 2026].

Tumors fail the invasion model quantitatively, and they do so consistently across tumor types. Volumetric doubling-time data assembled across common US cancers place measured tumor expansion in the range of 0.1–1.7% volume per day [Pienta, Brown, Amend, Gatenby 2026]. Translated to radial spread, even aggressive solid tumors advance at millimeters to centimeters per year. Breast cancer that grows at 70–140% per day in unrestricted culture grows at approximately 0.1–0.6% per day in patients. Prostate cancer volume doubling times are on the order of years. Even glioblastoma, the most aggressive common solid tumor, expands at approximately 1.4% volume per day [Pienta, Brown, Amend, Gatenby 2026; Stensjoen et al.; Lam et al.]. The intrinsic potential of cancer cells in culture exceeds realized tumor expansion by one to two orders of magnitude.

This is not a minor discrepancy. It is a structural mismatch that invalidates the invasion analogy as a quantitative model and, more importantly, misdirects therapy toward eradicating a moving front when the underlying ecology is one of bounded, internally regulated, locally engineered communities.

4.2 Why Invasion Analogies Fail All Three Tests

Classical invasion analogies fail three tests against observed tumor behavior [Pienta, Brown, Amend, Gatenby 2026]. First, front velocity in invasive species is set by intrinsic growth rate r and dispersal D , both realized close to potential in permissive habitat. Tumors have ample intrinsic growth rate but dramatically suppressed realized expansion — the habitat is not permissive; the tumor itself makes it impermissive through resource depletion and microenvironmental engineering. Second, invasive species spread is largely unbounded: limits are imposed externally by oceans, mountain ranges, or climate envelopes. Tumors hold sharp, imaging-defined boundaries that persist despite no geographic barrier; the boundary is maintained by the tumor's own internal ecology. Third, density behind the invasion front rapidly approaches the new carrying capacity. In tumors, the interior is resource-depleted, hypoxic, and acidic — the density gradient runs the wrong way, with the highest metabolic activity at the edge, not the interior.

4.3 The Prairie Dog Town as the Right Quantitative Model

Black-tailed prairie dogs (*Cynomys ludovicianus*) live in colonies known as towns: bounded, socially structured aggregations of burrow systems organized into coterie of 2–10 closely related individuals [Hoogland 1995]. Towns expand at approximately 27% per year in area and 30–40 meters of front per year in unconstrained settings — orders of magnitude below the species' intrinsic reproductive potential [Todd & Martin 2024; Milne-Laux & Sweitzer 2006]. Towns have sharply defined edges, depress local vegetation to bare soil through selective herbivory, engineer the landscape through burrow systems that recruit associated species (black-footed ferrets, burrowing owls, rattlesnakes, tiger salamanders), maintain a sophisticated alarm-call system that manages risk along town edges [Slobodchikoff et al. 1991], and crash episodically when sylvatic plague sweeps through [Eads & Biggins 2015; Cully & Williams 2001].

The prairie dog town is not an evocative metaphor for the tumor. It is a bounded, engineered, spatially structured ecological model of it, operating at the scale and with the structural features that actually govern tumor behavior. The case for the mapping rests on five specific structural homologies.

Bounded and edge-managed. Both systems hold sharp boundaries despite no external geographic barrier. Prairie dog towns have short, defended edges with coterie territories. Tumors hold imaging-defined boundaries with an active edge phenotype distinct from the interior [Lloyd et al. 2016]. In both cases, risk drives boundary dynamics: prairie dogs use alarm calls to manage predation risk at the edge; tumor cells at the edge face anoikis, immune surveillance, and the poorly perfused stroma — extreme risks that suppress cells venturing beyond the established boundary.

Habitat engineering. Prairie dogs are textbook ecosystem engineers: their burrow systems recruit a diverse associate community, alter soil chemistry and texture, and shift vegetation composition and productivity across the food web [Davidson et al. 2012; Duchardt et al. 2020]. Tumors engineer their microenvironment through angiogenesis, acidification of the ECM, matrix remodeling, and release of growth factors and cytokines that recruit and reprogram immune and stromal populations [Damaghi et al.]. In both cases, the community is not passively occupying its habitat; it is actively constructing it.

Keystone effects. Prairie dogs are paradigmatic keystone species: their ecological influence on North American grasslands is wildly disproportionate to their biomass [Davidson et al. 2012]. A tumor of a few grams can reshape the host's immune system, vasculature, microbiome, metabolism, neuroendocrine signaling, and behavior. Per gram, no other tissue rivals it. The disproportionate influence — keystone status — is a shared structural feature.

Resource-depleted interior with edge-interior phenotype partition. Prairie dog town interiors are resource-depleted; foraging is concentrated at the edge where vegetation is available but predation risk is higher. This food-safety tradeoff structures the spatial organization of prairie dog behavior [Lloyd et al. 2014]. The same tradeoff structures the tumor: cells at the edge face immune predation and anoikis but access nutrients; cells in the hypoxic acidic interior maximize proliferative resistance but at the cost of nutrient deprivation [Lloyd et al. 2014; Lloyd et al. 2016]. The edge-interior phenotype partition is the dominant spatial organizing axis of both systems.

Episodic crash under perturbation. Sylvatic plague (*Yersinia pestis*) can extirpate more than 90% of a prairie dog town in weeks, fragmenting colonies and reducing genetic variability [Eads & Biggins 2015; Cully & Williams 2001]. Chemotherapy and targeted therapy produce analogous episodic crashes in tumor cell populations, with partial recovery driven by the plastic, resistant subpopulations that survive the perturbation. In both cases, bounded, engineered communities have characteristic vulnerabilities that can be exploited by interventions targeting their internal structure rather than their geographic boundaries.

Figure 2 — [PLACEHOLDER] Three Styles of Spread and the Prairie-Dog Mapping. A three-panel conceptual figure: (A) Fisher-Skellam invasive-species expansion — a uniform front advancing into empty space, with density approaching carrying capacity behind the wave; (B) Prairie dog town expansion — bounded, internally structured, with resource-depressed interior, sharp edge, coterie organization, and slow realized expansion; (C) Solid tumor expansion — bounded, heterogeneous tissue with sharp edge, edge-interior phenotype partition, and realized growth far below intrinsic potential. The shared axes emphasize that tumors and prairie dog towns are bounded, internally structured communities rather than freely diffusing invasion waves. Adapted from [Pienta, Brown, Amend, Gatenby 2026].

4.4 The G-Function Formalization

To move from structural homology to quantitative prediction, the Cancer Ecology Center formalizes the town-as-tumor reading through the Vincent and Brown G-function: a fitness-generating function $G(v; u, x, E)$ that separates the heritable strategy v of a focal individual from the strategies u of all others it interacts with, their abundances x , and the environmental state E [Vincent & Brown 2005; Bukkuri et al. 2023].

The per-capita growth rate of a focal individual playing strategy v , embedded in population u with abundances x in environmental state E , is:

$$G(v; u, x, E) = (1/x_i)(dx_i/dt) \text{ evaluated at } v = u_i$$

Two coupled equations follow: an ecological dynamic, $dx_i/dt = x_i \cdot G(v; u, x, E) | \{v=u_i\}$, governing how population abundances change; and an evolutionary dynamic, du_i/dt proportional to $\partial G/\partial v | \{v=u_i\}$, governing how strategies evolve toward evolutionarily stable strategy (ESS) maxima [Vincent & Brown 2005]. G can be derived mechanistically from cell-level resource uptake, competition coefficients, public-goods production, motility costs, and therapy-induced selection [Bukkuri et al. 2023].

The G-function is the right framework for the town-as-tumor reading because it explicitly separates strategy from circumstance. Edge cells and interior cells share a strategy set but face different environmental states E ; coterie members and town-edge prairie dogs share a behavioral repertoire but face different predation risk levels. The same modeling object captures both. Critically, adaptive therapy can then be cast formally as a control problem in which the clinician modulates E — the environmental state — to steer the population composition toward a desired ecological equilibrium [Cunningham et al. 2018; Zhang et al. 2017].

An interactive G-function simulator implementing the town-as-tumor mapping is released as open source on GitHub at github.com/kpienta-cloud/Prairie-Dogs, with a live browser at kpienta-cloud.github.io/Prairie-Dogs/ [Pienta, Brown, Amend, Gatenby 2026].

4.5 Therapeutic Implication: Steer, Not Eradicate

The prairie-dog town reading generates a specific therapeutic imperative that is different from — and sometimes opposite to — the standard maximal cytoreduction approach. If the tumor is a bounded,

internally regulated ecological community rather than a freely diffusing invasion front, then the therapeutic goal is not to eradicate the moving front but to steer the community's internal ecology toward a state that is chronically suppressed, non-progressive, and compatible with sustained host survival.

The mechanism is ecological competition: sensitive cells, if maintained in sufficient numbers, ecologically suppress resistant subclones through competitive exclusion. Eradication of sensitive cells through aggressive continuous therapy removes this competitive suppression and creates ecological space for resistant subclones to expand. Maintaining a sensitive-cell reservoir — through treatment pauses, dose modulation, or adaptive scheduling — preserves the ecological competition that suppresses resistance [Gatenby et al. 2009; Zhang et al. 2017; Cunningham et al. 2018].

This is not a softer therapeutic approach. It is an ecologically informed one. The treatment goal shifts from maximal cytoreduction to bounded population-level control: keeping the tumor town stable rather than trying to drive the prairie dogs to extinction in a year.

4.6 Section Close

The prairie dog town is the right quantitative model for tumor behavior — not because of analogy, but because the structural features that govern realized tumor expansion (boundedness, edge management, habitat engineering, resource depletion, episodic crash) are the same structural features that govern the expansion of bounded ecological colonies in nature. The Fisher-Skellam invasion model is quantitatively wrong for tumors by one to two orders of magnitude and therapeutically misleading in directing effort toward eradicating a moving front. The town model is quantitatively consistent with measured tumor expansion rates and therapeutically productive in directing effort toward ecological steering of a bounded community.

Section 5. Forecast: The CEDT Layer

5.1 Why Existing Oncology Digital Twins Are Structurally Insufficient

The oncology digital twin literature has grown substantially over the past decade, clustering into three frameworks: biology-based mechanistic twins built on reaction-diffusion partial differential equations or logistic-growth ordinary differential equations; statistical virtual-patient matchers; and hybrid twins that merge mechanistic biology with AI surrogates and data-assimilation layers [CEDT v6.5, refs 2, 12, 13-19]. Together these efforts establish that an individualized, continuously updated, simulation-capable representation of cancer is technically feasible.

But nearly all published implementations share a common architectural failure: the tumor is represented as one or a small number of cell populations governed primarily by proliferation, migration, and therapy-induced death [CEDT v6.5, refs 14-19]. Patient physiology enters only as a constraint. Microenvironmental factors — vascularity, stroma, immune phenotype — are largely inferred from static parameters or population averages. The exposome — the cumulative record of carcinogenic, dietary, metabolic, and iatrogenic exposures that shapes both tumor origin and behavior — is largely absent [CEDT v6.5, refs 16-19].

This architecture mirrors the three blind spots identified in Section 1. It is structurally blind to the eco-evolutionary dynamics that drive therapy resistance; to immune-microenvironment state transitions among hot, excluded, and desert phenotypes; to spatial and resource-limited dynamics that govern adaptive scheduling and hypoxia-driven phenotype shifts; and to the exposome-tumor coupling across the life course. More fundamentally, most existing twins are calibrated once and run forward — they do not continuously update their estimate of the latent state from the data the clinical workflow already generates.

5.2 The CEDT State-Space Architecture

The Cancer Ecology Digital Twin addresses these failures in a single state-space framework. The latent state vector is:

$$\mathbf{X}(t) = [\mathbf{H}(t), \mathbf{E}(t), \mathbf{T}(t), \mathbf{M}(t), \mathbf{Q}(t)]^T$$

carrying five coupled subvectors [Pienta 2026]:

Host layer $H(t)$: organ reserve, systemic inflammation, immune competence, and clonal hematopoiesis. This subvector captures the patient's physiological capacity to tolerate treatment and to mount an immune response, and tracks its degradation over time and treatment exposure.

Exposure (exposome) layer $E(t)$: cumulative smoking, alcohol, and occupational exposures; isotope proxies from IsoMCED; and the pathway-level carcinogen burden scores computed by ExposoGraph (the E_{path} vector). This is the mathematical connection between the Exposure layer and the CEDT: the ExposoGraph output is not a separate risk score filed in a separate database; it is the $E(t)$ subvector of the CEDT latent state, continuously updated as new exposure or isotope data become available.

Tumor-intrinsic layer $T(t)$: clone abundances and fitness, driver genotype, genomic instability, plasticity score, resistance propensity, and metastatic hazard. In the current minimal executable subset, this comprises two clone abundances N_1 and N_2 (drug-sensitive and drug-resistant) with fitness rates r_1 and r_2 , scalar plasticity score ϕ , and resistance propensity ρ .

Tumor-ecosystem layer $M(t)$: anti-tumor and pro-tumor immune populations (N_{anti} , N_{pro}), stromal support S , vascular support V , habitat quality Q_{hab} , hypoxia burden H_{ac} , and the time-varying ecological interaction matrix $A(t)$ among all populations. The saturation parameter κ_{sat} governs predator-prey kinetics in the tumor-anti-tumor sector.

State-quality layer $Q(t)$: posterior covariance, missingness mask, observation confidence, and time since last update. This layer explicitly represents what is known and how confidently, enabling the CEDT to degrade its forecast uncertainty appropriately when observations are missing or delayed — the clinical reality of oncology care.

5.3 The Dynamical Core: Generalized Lotka-Volterra with Resource Coupling

The CEDT's dynamical core is a generalized Lotka-Volterra (GLV) system coupled to explicit resource and habitat-quality compartments, with state-dependent growth-rate functions that encode hypoxia sensitivity, drug pharmacodynamics, and plasticity-driven phenotype switching [Pienta 2026]. The interaction matrix $A(t)$ is treated as a

latent quantity — not a fixed parameter estimated once at calibration, but a time-varying quantity that is sequentially re-estimated from sparse, irregular, multi-modal clinical observations using Bayesian data assimilation.

Three filter architectures are implemented: the Extended Kalman Filter (EKF), which linearizes the nonlinear GLV system at each assimilation step; the Ensemble Kalman Filter (EnKF), which propagates an ensemble of state realizations and avoids linearization; and the Unscented Kalman Filter (UKF), which uses sigma points to capture higher-order moments of the nonlinear distribution [CEDT v6.5, refs 53-55]. The choice among filters depends on the degree of nonlinearity in the current trajectory and is a parameter of the CEDT deployment.

Why each layer of $X(t)$ is necessary can be stated precisely. T alone is logistic growth — a model that ignores all of the ecology that drives resistance. $T + M$ is ecology — tumor-immune-stroma dynamics that produce the competitive exclusion, coexistence, and cooperative outgrowth regimes that govern therapeutic response. Adding H gives organ reserve and systemic inflammation, which gate drug delivery and immune capacity. Adding E couples the model to exposure history, so that the CEDT “knows” that this patient’s tumor originated in a high-carcinogen-flux, glutathione-depleted context and that the carcinogen-driven mutational signature affects resistance propensity. Adding Q gives observation-confidence handling — the mechanism by which the CEDT remains calibrated in the clinical reality of sparse, irregular, missing data.

5.4 In Silico Results: Eco-Evolutionary Regimes and Adaptive Therapy

Evaluated on cohorts of 1,000 simulated virtual patients with interaction matrix A sampled across documented phase boundaries, the CEDT reproduces canonical eco-evolutionary regimes — competitive exclusion, stable coexistence, and cooperative outgrowth — and identifies critical transitions in the interaction-matrix spectrum consistent with prior analytical results [Pienta 2026]. Sweeping clonal diversity from one to ten co-existing clones, the CEDT reproduces a sharp transition from immune-controlled dormancy to immune escape at a diversity threshold tracking $r_{\text{kill}} / (\kappa_{\text{sat}} \cdot D)$, where D is effective clone diversity — consistent with prior tumor-immune predator-prey analyses [CEDT v6.5, refs 29, 30].

The CEDT also reproduces three resource-depletion transients that standard static-coefficient GLV cannot generate: hypoxia-driven phenotype shift, antiangiogenic rebound, and treatment-cycle resource recovery [Pienta 2026]. These transients are generated by the explicit $V(t)$ and $Q_{\text{hab}}(t)$ compartments coupled to growth rates through the hypoxia-sensitivity function $\phi_i(Q_{\text{hab}})$; they emerge from the model dynamics rather than from programmed parameters.

For adaptive versus continuous therapy, each of 1,000 virtual patients was forecast forward 24 months under continuous standard dosing and under adaptive dosing with a 50% burden-trigger pause-resume policy. Predicted median time to progression was 9.4 months (IQR 5.1–16.3) under continuous dosing and 12.8 months (IQR 6.7–21.4) under adaptive dosing [Pienta 2026]. Per-patient preference depended on three features: resistant-clone fraction; relative fitness cost of resistance; and plasticity score. Plasticity reverses the sign of adaptive benefit — at high plasticity ϕ approximately 0.55, phenotype switching compresses the competitive-release window and eliminates adaptive therapy benefit — consistent with predictions that fixed-identity GLV models overestimate adaptive benefit relative to plasticity-aware models [CEDT v6.5, refs 76-79].

5.5 Identifiability: The Two Practical Anchors

A common and legitimate critique of GLV models is that the ecological interaction matrix A is underdetermined: with n populations, A has n^2 entries, and recovering all of them from sparse time-series data requires stronger identifiability assumptions than most clinical datasets can support. The CEDT addresses this through two practical design choices that jointly recover identifiability for most interaction coefficients under clinically realistic data conditions [Pienta 2026].

The first anchor is a single absolute cell-count measurement at baseline: imaging volume paired with pathology cellularity converts 7 of 9 marginally identified interaction coefficients into the well-identified class [Pienta 2026]. This anchors the absolute-abundance scale and resolves the scaling degeneracy that leaves relative-abundance time series underdetermined.

The second anchor is functional-group aggregation: aggregating the five-population model into four functional guilds — anti-tumor immune, pro-tumor immune, tumor-sensitive, and tumor-resistant — reduces the interaction matrix from 25 to 16 entries, of which 14

become well-identified under the baseline clinical schedule with no additional data [Pienta 2026]. Forecast performance under aggregation is within 3 percentage points of the full-resolution model.

Under clinically realistic sparse conditions (imaging every 8–12 weeks, ctDNA every 4–6 weeks, tissue biopsy at baseline and progression, laboratory panels every 2–4 weeks, approximately 30% missing-at-random dropout), 10 of 25 interaction coefficients in the five-population model are well-identified from standard clinical channels [Pienta 2026]. ctDNA is the most information-dense observation channel per unit cost: removing it increases the 12-month forecast credible-interval width by a factor of 2.4, compared to 1.7 for halving imaging frequency and 1.2 for removing tissue sequencing beyond baseline.

5.6 Falsifiability and the Retrospective Bruchovsky Protocol

Three explicit empirical predictions on prospective intermittent-therapy cohorts would falsify CEDT v1 if violated [Pienta 2026]:

1. CEDT will predict individual patient time-to-progression on an intermittent-therapy prostate cancer cohort with mean absolute error less than a pre-specified threshold, beating five comparator models (standard logistic growth, PK/PD, static GLV, best-fit statistical matcher, population pharmacokinetic model).
2. The plasticity threshold $\phi \approx 0.55$ at which adaptive therapy loses benefit, estimated from the virtual-patient simulator, will be validated against an independent cohort in which adaptive therapy was applied across a phenotypic spectrum.
3. The CEDT-estimated ecological interaction matrix $A(t)$ will identify regime-shift events (coexistence to competitive exclusion; immune control to immune escape) at least one clinical observation interval before they manifest as PSA progression or imaging-defined tumor growth.

The prospective validation path runs through the Bruchovsky et al. intermittent-androgen-deprivation prostate cancer cohort as the primary retrospective-fit benchmark [CEDT v6.5, ref 131]. A fully specified retrospective-fit protocol with a head-to-head benchmark template against five comparator models is included in the CEDT manuscript [Pienta 2026].

Figure 3 — [PLACEHOLDER] CEDT State-Space Architecture and Forecast-Update Loop. A three-panel figure: (A) The five-layer latent state $X(t) = [H, E, T, M, Q]$ with component populations labeled; (B) The dynamical core — GLV system with resource compartments $V(t)$ and $Q_{\text{hab}}(t)$, and the EKF/EnKF/UKF assimilation filter; (C) The forecast-update loop over a 24-month patient trajectory, showing prior forecast, observation-driven update, and posterior state evolution with credible intervals. The figure emphasizes that the CEDT does not calibrate once and run forward; it continuously revises its state estimate as new clinical data arrive. Source: [Pienta 2026].

5.7 Section Close

The CEDT is the only oncology digital twin architecture that takes both the exposome and the tumor ecosystem seriously as first-class state variables, and the only one designed for the data sparsity reality of the oncology clinic. Its failure-mode stress tests against the seven documented GLV failure modes, its Sobol global sensitivity analysis over all non-assimilated parameters, and its explicit identifiability protocol distinguish it from the model confidence theater that characterizes much of the digital twin literature. It is positioned as a falsifiable proof-of-concept architecture, not a clinical tool — but with a fully specified path to prospective validation.

Section 6. The Unified Program: How the Three Layers Compose

6.1 Mathematical Composition, Not Rhetorical Juxtaposition

The three layers of the Cancer Ecology Center program — Exposure, Ecosystem, and Assimilation — are not parallel research tracks that illuminate the same disease from different angles. They are mathematically coupled, with explicit quantitative connections that make the integration computable rather than merely rhetorical.

Connection 1: Exposograph → CEDT. The carcinogen-flux burden vector E_{path} , computed by ExposoGraph 2.0 from a patient's exposure history and genotype, is literally the $E(t)$ subvector of the CEDT latent state. When a patient's NHANES-derived occupational exposure data and CYP/NAT genotyping are fed into ExposoGraph, the output is a

tissue-specific carcinogen burden score that populates the $E(t)$ compartment of the CEDT at baseline. As the patient accumulates additional exposure history over the follow-up period, $E(t)$ is updated accordingly. The IsoMCED isotope readout provides a non-invasive longitudinal update channel for $E(t)$ and $H(t)$ simultaneously — the hair isotope ratios reflect both the exposure burden and the host metabolic response to it. This coupling means that the CEDT does not assume a generic “smoker” or “non-smoker” exposure state; it ingests the actual per-person, per-tissue, per-carcinogen-class flux vector computed from the patient’s individual exposure history and genetic background.

Connection 2: Prairie-Dog/G-function → CEDT. The behavioral ecology of the tumor — bounded, internally regulated, resource-depleting, edge-managed — is not merely an analogy that motivates the CEDT’s structure. It is the biological rationale for specific architectural choices. The GLV core is chosen because bounded, resource-competing ecological communities are the correct physical model for the tumor ecosystem, as demonstrated by the quantitative comparison of tumor expansion rates with ecological invasion speeds in Section 4. The resource compartments $V(t)$ and $Q_{\text{hab}}(t)$ are present because the tumor depletes oxygen and nutrients in an internally structured way, exactly as prairie dog towns deplete local vegetation. The carrying capacity structure and the edge-interior phenotype partition formalized by the Lloyd et al. food-safety tradeoff are encoded in the $M(t)$ subvector. The bounded expansion dynamics of the G-function are the reason that GLV with a finite carrying capacity — not an open-ended reaction-diffusion model — is the right dynamical core.

Connection 3: G-function → Adaptive Therapy Control Law. The G-function formalizes adaptive therapy as a control problem in which the clinician modulates the environmental state E to steer the tumor population composition. The CEDT individualizes this control law: it estimates the current population composition (resistant-clone fraction, plasticity score, immune phenotype) from the patient’s clinical data, predicts the trajectory under each candidate adaptive therapy protocol, and identifies the duty cycle and burden trigger that maximizes time to progression for this patient. The control law is not applied from a table; it is computed in silico for each patient at each clinical visit. The G-function provides the theoretical framework; the CEDT provides the patient-specific implementation.

Connection 4: IsoMCED → Population-Scale E+H Monitoring. At the individual level, IsoMCED provides non-invasive longitudinal updates to the E(t) and H(t) subvectors of the CEDT. At the population level, it enables cancer screening in settings where cfDNA-based multi-cancer early detection tests are not accessible due to cost or cold-chain requirements. The sub-\$20 per sample cost and absence of cold-chain requirements make IsoMCED deployable in low- and middle-income country populations that carry a disproportionate share of the projected 2050 cancer mortality burden. Population-scale IsoMCED screening generates E_path data at the cohort level that feeds back into ExposoGraph model validation against All of Us and NHANES datasets.

6.2 Three Layered Falsifiable Predictions

The Cancer Ecology Center program is not a unified theory of cancer in the sense of a grand claim that it explains everything. It is an integration architecture for three specific layers, and it makes three specific falsifiable predictions — one per layer — that together constitute the empirical test of the program [canonical_doc_brief.md; Pienta 2026].

Prediction 1 (Exposograph layer): Multi-carcinogen synergy scores computed by ExposoGraph 2.0 will predict tissue-specific cancer incidence in All of Us and NHANES cohorts with significantly better fit than single-carcinogen models, with effect size and direction pre-registered before analysis. The specific pre-registered hypothesis is that individuals with ExposoGraph-computed synergy scores above the 75th percentile will have tissue-specific cancer incidence rates at least 1.5-fold higher than individuals below the 25th percentile, controlling for age, sex, and population ancestry group.

Prediction 2 (Prairie-Dog/G-function layer): Maintaining a sensitive-cell reservoir through adaptive therapy protocols informed by the G-function framework will delay time to progression compared to standard-of-care continuous dosing in at least one additional tumor type beyond prostate cancer, replicating the Zhang et al. 2017 result in a second tumor type with prospective design and pre-registered adaptive protocol. The mechanism — competitive suppression of resistant subclones by maintained sensitive-cell populations — must be confirmable through longitudinal tumor heterogeneity assessment.

Prediction 3 (CEDT layer): Prospective forecast accuracy on an intermittent-therapy prostate cancer cohort will exceed five comparator models (standard logistic growth, PK/PD, static GLV, best-fit statistical matcher, population pharmacokinetic model) by a pre-

specified margin on the primary endpoint of 12-month time-to-progression mean absolute error. The margin, endpoint, and cohort definition are fully specified in the retrospective Bruchovsky fit protocol [Pienta 2026].

6.3 The Architecture Is Open Science

Every component of the Cancer Ecology Center program — Exposograph, GeoToxGraph, CEDT, Prairie-Dogs G-function simulator, IsoMCED model weights — is released or will be released as open source under MIT or CC-BY licensing, with Zenodo DOIs for dataset versions and GitHub repositories for code [canonical_doc_brief.md]. This is not a courtesy gesture. It is a scientific requirement: falsifiability demands transparency, reproducibility requires code availability, and the 16-institution Cancer Ecology Consortium requires shared computational infrastructure. The open-science architecture is the prerequisite for the collaborative validation program.

Figure 4 — [PLACEHOLDER] Cancer Ecology Center Program Integration Diagram. A single-page schematic showing the three-layer architecture with explicit mathematical connections. Left column: Exposure layer (ExposoGraph $\rightarrow$ E_{path} , GeoToxGraph $\rightarrow$ contrast map, IsoMCED $\rightarrow$ $\delta^{13}\text{C}/^{15}\text{N}/^{34}\text{S}$). Center column: CEDT latent state $X(t) = [H, E, T, M, Q]$ with arrows showing which instrument populates which subvector. Right column: Ecosystem layer (Prairie-Dog G-function $\rightarrow$ GLV core architecture, adaptive therapy control law). Bottom: Three falsifiable predictions, one per layer, with pre-registration status indicated. Caption: The three layers compose mathematically, not just conceptually. The Exposure layer output populates the $E(t)$ subvector of CEDT; the Ecosystem behavioral model is the biological rationale for the GLV core; the G-function is the control law that CEDT individualizes for each patient.

Section 7. Critics, Objections, and Boundaries of the Claim

7.1 “This Is Over-Claiming”

The most important objection to acknowledge is the broadest: that a research program spanning carcinogenomics, cancer ecology, evolutionary game theory, digital twins, and stable isotope chemistry is necessarily over-claiming, conflating metaphor with mechanism, or assembling a rhetorical edifice that outpaces its empirical foundations.

The response is architectural: the Cancer Ecology Center program is not claiming a unified theory of cancer. It is claiming an integration architecture for three specific, independently publishable, independently falsifiable research layers. ExposoGraph 2.0 publishes its own quantitative predictions about multi-carcinogen synergy and genotype-specific risk, and those predictions can be falsified independently of the CEDT and independently of the prairie-dog reading. The Prairie-Dog Perspective publishes its own quantitative comparison of tumor and ecological expansion rates, and that comparison is falsifiable independently of the exposome work. The CEDT publishes its own three explicit falsifiable predictions on prospective intermittent-therapy cohorts.

The integration claim — that these three layers compose into a coherent program — is an architectural claim about how the outputs of one layer feed the inputs of another. That claim is falsifiable: if the ExposoGraph E_{path} vector provides no predictive value for CEDT forecast accuracy beyond what is captured by smoking history alone, the integration claim fails for the Exposure → Assimilation connection. The program is not immune to falsification; it is densely exposed to it.

7.2 “GLV Models Are Notoriously Underdetermined”

This is a legitimate scientific concern and the CEDT addresses it directly. The interaction matrix A for an n -population system has n^2 entries. With $n = 5$ populations and standard clinical observation schedules, recovering all 25 entries from sparse longitudinal data is not generically possible without additional constraints. The seven documented failure modes of standard generalized Lotka-Volterra — including negative equilibria, sensitivity to initial conditions,

extinction-order dependence on initial abundances, and parameter non-identifiability — are stress-tested explicitly in the CEDT failure-mode analysis [Pienta 2026].

The CEDT's response to underdetermination is the two-anchor strategy described in Section 5.5: one absolute cell-count measurement at baseline, and functional-group aggregation. Together these recover identifiability for 14 of 16 interaction coefficients in the four-group aggregated model under standard clinical schedules [Pienta 2026]. The residual 2 weakly identified coefficients (stroma-vasculature and stroma-stroma self-regulation) require spatial pathology data rarely available clinically and are flagged in the Q(t) state-quality layer accordingly — the CEDT does not pretend to know what it cannot know.

The Sobol global sensitivity analysis over the eight fixed parameters not subject to assimilation confirms that absolute-abundance scaling dominates forecast uncertainty (first-order Sobol index $S_i = 0.95$ for time-to-progression, $S_i = 1.00$ for 12-month log-PSA), while all seven other fixed parameters show first-order indices below $|0.12|$ [Pienta 2026]. This means that the quantitative anchoring of the absolute tumor cell count — the most important single data point for reducing forecast uncertainty — is also the most clinically controllable intervention: it requires only digitally quantified H&E cellularity paired with imaging volume at baseline.

7.3 “The Exposome Is a Measurement Problem, Not a Model Problem”

This objection is partly correct and partly a false dichotomy. It is correct that the exposome has historically been a measurement problem: without dense longitudinal exposure assessment data linked to biological endpoints, the exposome concept is rich but the epidemiology is weak. The first-generation ExposoGraph was criticized for exactly this: it provided a qualitative visualization of carcinogen-metabolism pathway topology without quantitative exposure inputs.

ExposoGraph 2.0 addresses the measurement problem directly by integrating NHANES, EPA IRIS, and OSHA surveillance data as quantified exposure inputs, and by pairing these with GTEx v8 tissue-specific expression data and gnomAD v4.1 population-scale genotype distributions [Kazi, Sarigiannis, Pienta 2026]. The result is a platform

that can compute tissue-specific risk scores from real-world exposure surveillance data, not just from hypothetical maximum-exposure scenarios.

IsoMCED addresses the measurement problem at a different level: by reading the biological output of cumulative exposure in hair keratin, it sidesteps the requirement for complete exposure history reconstruction. The isotope record integrates exposure, host metabolism, and tumor metabolism in a single sample. This is why IsoMCED and ExposoGraph are complementary rather than redundant: ExposoGraph computes forward from exposure to predicted risk; IsoMCED reads backward from biology to inferred cumulative metabolic state.

The objection that this is “a measurement problem, not a model problem” implies that better measurement would make the model unnecessary. This is wrong: better measurement without a quantitative model for how exposures interact across tissues and genotypes remains descriptive epidemiology. The model is what enables prediction for an individual from population-scale measurement data.

7.4 “Adaptive Therapy Isn’t Generalizable Beyond Prostate Cancer”

The Zhang et al. 2017 intermittent abiraterone trial in metastatic castration-sensitive prostate cancer remains the most rigorous prospective demonstration of evolutionary steering in a clinical setting [Zhang et al. 2017]. The Cancer Ecology Center’s Prediction 2 (Section 6.2) acknowledges that generalizability requires prospective demonstration in a second tumor type. This is not a weakness of the program; it is the program.

The growing evidence base for adaptive therapy extends to breast cancer (Gatenby et al., small cohort), glioblastoma (Cunningham et al., computational demonstration), and melanoma (computational models). None of these is a randomized controlled trial. The CEDT is designed to identify which patients in which tumor types are most likely to benefit from adaptive scheduling — stratified by resistant-clone fraction, plasticity score, and immune phenotype — and to design adaptive protocols individualized to each patient rather than applying a single threshold rule population-wide. The combination of

patient stratification (who benefits?) and protocol individualization (at what trigger?) is what distinguishes the CEDT-informed approach from the single-protocol adaptive therapy trials published to date.

7.5 “Digital Twins Haven’t Delivered Clinically”

This objection reflects an accurate assessment of the current state of oncology digital twins: none has yet demonstrably improved individual patient outcomes in a randomized clinical trial. The Cancer Ecology Center does not claim otherwise. The CEDT is positioned explicitly as a falsifiable proof-of-concept architecture, not a clinical tool.

The distinction between the CEDT and existing twins is the state architecture. A twin that models the tumor as a single growing object cannot forecast resistance, because resistance is an ecological process that requires modeling the interaction between sensitive and resistant subclones, the immune compartment, and the resource landscape. A twin without an exposure substate cannot account for the carcinogen-driven mutational signature that shapes resistance propensity. A twin without Bayesian data assimilation cannot update its forecast when clinical observations disagree with the prior trajectory. The CEDT corrects all three. Whether correcting these architectural failures translates to improved clinical outcomes is exactly what the prospective validation program will determine — and the three falsifiable predictions in Section 6.2 are the test.

7.6 What This Program Does Not Claim

Clarity about boundaries is a scientific obligation. The Cancer Ecology Center program does not claim:

- A cure framework. Steering is not curing. The goal of adaptive therapy is extended progression-free survival and preservation of quality of life, not permanent eradication.
- A universal model of cancer. Hematologic malignancies, which are not solid tumor-forming communities, require different ecological mappings. Some aggressive metastatic cancers may not be steerable by current adaptive therapy protocols. The program addresses solid tumor ecology in detail and acknowledges that extension to other disease contexts requires separate work.

- A replacement for molecular biology or clinical pathology. ExposoGraph depends on the mechanistic enzymology and pharmacogenomics literature built by the molecular biology community. The CEDT's $T(t)$ subvector depends on genomic and transcriptomic characterization of tumor subclones. The Prairie-Dog mapping requires quantitative tumor pathology for the edge-interior phenotype partition. The Cancer Ecology Center program is an integration layer atop the molecular, cellular, and clinical foundations of oncology — not a replacement for them.
- Complete parameter identifiability. The CEDT's identifiability analysis is explicit that 6 of 25 interaction coefficients in the five-population model are weakly identified under standard clinical schedules. These weaknesses are tracked in the $Q(t)$ state-quality layer, not hidden.

Acknowledging these boundaries is part of what distinguishes a scientifically serious integration architecture from a grand theory program that will eventually collapse under the weight of its own claims.

Section 8. What This Means for the Clinic: An End-to-End Reframe

8.1 From Population-Average Workflows to Individualized Ecological Reading

The Cancer Ecology Center program implies a specific reframe of clinical practice at every stage of the oncology workflow: risk assessment, screening, diagnosis, treatment planning, and surveillance. This reframe is not incremental — it does not propose adding one new biomarker or one new decision tool to an otherwise unchanged workflow. It proposes a different architecture: one in which the patient's exposure history, the tumor's ecological structure, and the evolving model of disease behavior are all first-class inputs to every clinical decision.

8.2 Risk: From Checkboxes to Personal Risk Landscapes

Current risk assessment tools for cancer — clinical guidelines for screening eligibility, genetic counseling referral criteria, primary prevention recommendations — are built on population-level relative

risks stratified by a small number of binary or categorical variables: family history (positive/negative), tobacco (ever/never or pack-years), age (above/below threshold). These tools efficiently triage large populations but provide no individualized quantitative forecast of absolute risk for a specific person with specific exposures and specific genotype.

The ExposoGraph risk engine transforms this. Given a patient's exposure history (from NHANES-format questionnaire data or occupational hygiene records), CYP/NAT/GST/ALDH/SULT/UGT genotype (from clinical pharmacogenomics panel or research-grade whole-exome sequencing), and tissue of interest, ExposoGraph computes a tissue-specific carcinogen flux burden score that integrates all exposures, their synergistic interactions, and their genotype-specific processing rates. The result is not a relative risk referenced to never-smokers; it is an absolute flux estimate for this person's carcinogen processing capacity in the tissue of concern [Kazi, Sarigiannis, Pienta 2026].

At population scale, IsoMCED enables the first genuinely deployable instrument for multi-cancer risk stratification that does not require venipuncture or cold chain. A hair sample collected by a community health worker, shipped at ambient temperature, and processed at an IRMS facility for under \$20 generates a tri-isotope signature that, when decoded by the IsoMCED classifier, flags individuals who have undergone the carbon, nitrogen, and sulfur metabolic shifts associated with cancer [Pienta Fast Grant 2026]. This is not a diagnostic test; it is a risk-stratification instrument that identifies individuals warranting further evaluation. Its cost-equity properties are the inverse of cfDNA-based multi-cancer early detection: it is cheaper, more deployable, and potentially more sensitive to the cumulative metabolic effects of multi-year carcinogen exposure than snapshot liquid biopsy.

8.3 Screening: Hair Isotopes as a Population-Scale Instrument

Multi-cancer early detection through cfDNA methylation or protein biomarker panels has demonstrated clinical utility but costs \$250–\$1,000 per test and requires cold chain from phlebotomy to laboratory — a requirement that makes population-scale deployment in low- and middle-income countries practically infeasible [Pienta Fast Grant 2026]. IsoMCED's sub-\$20 per sample cost and ambient-temperature stability close this access gap.

The planned IsoMCED validation program — 600 archived samples across six cancer types, gradient-boosted and transformer classifiers, CE-IVDR and WHO PQDx pre-registration — establishes the analytical validation evidence base required for regulatory qualification [Pienta Fast Grant 2026]. Commercial translation through Kreftect, Inc. provides the CAP/CLIA pathway for clinical deployment in the United States market, while the WHO PQDx pathway enables deployment through national health programs in low- and middle-income countries.

8.4 Diagnosis: Tumor Ecology Classification Supplements TNM/Grade

The current staging system — TNM (tumor size/nodal status/metastasis) supplemented by histological grade and selected biomarker panels — classifies tumors by their gross anatomical extent and cellular appearance. It does not classify tumors by their ecological structure: the edge-interior phenotype partition, the immune microenvironment phenotype (hot/cold/excluded), the resource landscape (well-vascularized versus hypoxic interior), or the exposure-history fingerprint encoded in the mutational signature.

The Cancer Ecology Center program proposes that tumor ecology classification — standardized characterization of these ecological parameters from standard-of-care imaging, pathology, and genomics data — should supplement TNM/grade as a prognostic and predictive instrument. An imaging-derived edge-to-interior metabolic gradient (from FDG-PET or perfusion MRI), a digital pathology-derived edge-interior phenotype partition score (from H&E or multiplex immunofluorescence), an immune phenotype classification (from tumor-infiltrating lymphocyte density and spatial organization), and an ExposoGraph-derived exposure-history fingerprint (from mutational signature deconvolution) together define an ecological profile that predicts adaptive therapy benefit with specificity that TNM staging alone cannot.

8.5 Treatment Planning: G-Function-Informed Adaptive Therapy as Default

The standard treatment-planning workflow for metastatic cancer selects a regimen from evidence-based guidelines — protocols derived from the average patient in a trial population — and applies it continuously until progression. This workflow is not individualized; it is population-average applied to one patient.

G-function-informed adaptive therapy replaces this with an ecologically informed control problem. The CEDT estimates the current tumor ecological state (resistant-clone fraction, plasticity score, immune phenotype) from the patient's available clinical data. It simulates the tumor trajectory under each candidate adaptive protocol (different burden triggers, duty cycles, dose levels) and identifies the protocol that maximizes predicted time to progression for this patient, accounting for uncertainty in the estimated ecological state. The treatment plan is not selected from a table; it is computed *in silico* for this patient at this time in their disease course.

This is not a distant aspiration. The computational infrastructure — the CEDT, the G-function simulator, the observation-channel mapping — is built. What is missing is the prospective clinical validation program that tests whether CEDT-individualized adaptive therapy protocols produce better outcomes than standard adaptive therapy protocols in randomized or adaptive clinical trial designs. That program is the clinical translation path.

8.6 Surveillance: CEDT as a Continuous Ecological Monitor

Standard oncologic surveillance is event-driven: imaging is performed on a fixed schedule (every 3 months, every 6 months), and the result is compared to the prior scan. The comparison is anatomical — did the tumor grow? — and categorical — progression, stable, partial response, complete response. The clinical response is also categorical: if progression, change regimen.

The CEDT enables a different surveillance architecture. The CEDT state estimate is updated at every clinical observation — every lab result, every imaging study, every ctDNA assay — and propagated forward between observations with the GLV dynamical model. The result is a continuously updated probabilistic forecast of the tumor trajectory: not just “did it grow?” but “where is it heading, at what rate, and what

does the ecological state suggest about imminent regime shifts?” The CEDT flags ecological regime shifts — transitions from coexistence to competitive exclusion, from immune control to immune escape — before they manifest as clinical progression events, enabling intervention at the earliest ecologically detectable moment rather than after anatomical progression has already occurred.

Section 9. The 10-Year Research Program

9.1 Cohorts and Clinical Platforms

The Cancer Ecology Center’s empirical foundation rests on three cohort platforms. The Hopkins Biobank for Genomics and Exposomics at Hopkins (BGES) is the primary internal platform: currently enrolling at the Brady Urological Institute, with an R01 in progress to expand to 1,000 participants with deep exposure assessment (NHANES-format questionnaires, occupational hygiene records, pharmacogenomics panels), hair isotope collection (IsoMCED), longitudinal imaging, and ctDNA surveillance. The BGES is designed to provide all observation modalities required to populate the full CEDT latent state vector for each enrolled participant.

The All of Us Research Program provides the population-scale validation cohort for ExposoGraph predictions: over 500,000 participants with survey data, electronic health records, biobanked samples, and genomic data. The All of Us adapter module in ExposoGraph 2.0 enables direct mapping of All of Us survey items to NHANES-format exposure variables, allowing the ExposoGraph risk engine to generate tissue-specific risk scores for the All of Us cohort and test the pre-registered Prediction 1 against actual cancer incidence outcomes [Kazi, Sarigiannis, Pienta 2026].

International partnerships through the Multi-elemental Insights of Origins (MIO) Consortium — currently including Lund University, Aristotle University of Thessaloniki, the University of Copenhagen, and the Cancer Ecology Center — provide European cohort data for ExposoGraph validation against HBM4EU/PARC harmonized biomonitoring data and for GeoToxGraph extension to the European industrial chemical exposure landscape.

9.2 Instruments and Publications

The four primary Cancer Ecology Center instruments have specific near-term publication and validation milestones. ExposoGraph 2.0 is under review at PLOS Computational Biology. The CEDT manuscript (v6.5) is in final preparation for submission. The Prairie-Dog Perspective is in revision, with the interactive G-function simulator released at kpienta-cloud.github.io/Prairie-Dogs/. The GeoToxGraph is released as an open-source browser at kpienta-cloud.github.io/geobacter/, with the evolutionary contrast mapping manuscript in revision.

IsoMCED has pilot funding through the Biswas Family Foundation Fast Grant (Spring 2026) for the 600-sample analytical validation study, with Krefect, Inc. providing the commercial translation pathway to CAP/CLIA qualification and CE-IVDR regulatory submission. The CarcinoGenomics R01 (Research Strategy v61) is in active NIH submission, targeting funding of the expanded BGES cohort and the prospective CEDT validation study.

9.3 Validation Program

Three falsifiable predictions (Section 6.2) define the validation program, each with a specific validation design. Prediction 1 (ExposoGraph) will be tested in the All of Us cohort with pre-registered hypotheses lodged in ClinicalTrials.gov or OSF before analysis. Prediction 2 (Adaptive Therapy) will be tested in a prospective adaptive therapy cohort in a second tumor type, with pre-specified adaptive protocol, burden trigger, and primary endpoint of time to progression. Prediction 3 (CEDT) will be tested retrospectively against the Bruchovsky intermittent-androgen-deprivation cohort as the benchmark fit protocol, followed by a prospective forecasting study on a new intermittent-therapy prostate cancer cohort.

The validation program is designed to be sequentially falsifiable: each prediction can fail independently of the others. Failure of Prediction 1 would falsify the ExposoGraph layer without falsifying the CEDT or the adaptive therapy program. Failure of Prediction 3 would require CEDT revision without necessarily implicating the ExposoGraph or adaptive therapy programs. This independence is a scientific virtue, not a hedge: it means that the program can learn and revise from failures rather than requiring that every prediction succeed simultaneously.

9.4 The Cancer Ecology Consortium

The 16-institution Cancer Ecology Consortium operationalizes the Cancer Ecology Center program at the scale required for prospective validation and international cohort analysis. Member institutions contribute patient cohorts, computational infrastructure, or scientific expertise across the three program layers. The Consortium's governance structure, data sharing agreements, and publication policies are organized around the open-science principle: all data, code, and models produced through Consortium activities are released under MIT/CC-BY licensing with Zenodo DOIs, enabling independent replication and extension by the broader cancer research community.

9.5 Open Science Architecture

The Cancer Ecology Center's commitment to open science is operationalized through specific releases: ExposoGraph code and data at github.com/kpienta-cloud/Exposograph; GeoToxGraph code and data at github.com/kpienta-cloud/geobacter; CEDT code at github.com/kpienta-cloud/CEDT; Prairie-Dogs simulator at github.com/kpienta-cloud/Prairie-Dogs; IsoMCED model weights, training code, and de-identified isotope spectra released on Zenodo upon publication. Every publication from the program submits to preprint servers (bioRxiv or medRxiv) simultaneously with journal submission.

This is not altruism. Open science is the fastest path to the validation program. The Cancer Ecology Center cannot alone conduct all the prospective cohort studies required to validate the three predictions across tumor types, ancestry groups, and healthcare settings. Making the tools available to external collaborators at zero cost and with full documentation — including working tutorial notebooks for each component — is the mechanism by which the Cancer Ecology Consortium grows and the validation program accelerates.

Section 10. Closing: A New Compact

10.1 What We Owe Patients

Every year approximately 10 million people die of cancer [CEDT v6.5, ref 136]. Most of them receive treatment decisions made on the basis of what has worked on average for patients who superficially resemble them. Their individual exposure history — the combination of

chemicals, the genotype-specific metabolic processing, the tissue-specific carcinogen flux that preceded their cancer — is not in the model. The ecological dynamics of their tumor — the balance of sensitive and resistant subclones, the immune microenvironment state, the resource landscape that determines which therapeutic strategies will select for resistance — is not in the model. The longitudinal clinical data generated by their surveillance — the imaging, the ctDNA, the laboratory panels — is used to assess response to the current regimen but is not integrated into a continuously updating model of where the disease is heading next.

We owe these patients more than population-average decisions. We owe them a clinical decision architecture that takes their individual exposure history seriously, models their tumor as a functioning ecosystem, and continuously updates its forecast as new data arrive. The Cancer Ecology Center program is built to deliver that architecture.

10.2 What We Owe Science

The molecular revolution in cancer biology produced an extraordinary catalog of oncogenes, tumor suppressors, signaling pathways, mutational signatures, and molecular subtypes. It did not produce a quantitative, computable, falsifiable framework for predicting what will happen to an individual patient under each of several candidate treatment paths. The reason is architectural: molecular catalogs are not dynamical models. Knowing that a patient has a KRAS G12C mutation tells a clinician which drug to try; it does not tell the clinician how long it will work, when resistance will emerge, what ecological mechanism will drive that resistance, or what adaptive protocol would delay it.

We owe science more than incremental single-gene and single-chemical studies that add to the catalog without building the dynamic model. The Cancer Ecology Center program does not propose to replace molecular biology; it proposes to build the integration layer that translates the molecular catalog into a dynamical, ecological, data-assimilating model of individual cancer behavior.

10.3 What We Owe the Field

The cancer research field has accumulated, over decades, the intellectual components needed for the integration program described in this white paper: evolutionary theory from Nowell and Tomlinson;

game-theoretic formalism from Vincent and Brown; ecological frameworks from Gatenby, Dujon, and Aktipis; adaptive therapy clinical evidence from Zhang et al.; exposome concepts from Wild; clonal architecture from Maley, Greaves, and Polyak; pharmacogenomics from the CPIC community. Each of these contributions was necessary. None alone is sufficient.

We owe the field a falsifiable, computable, integrative framework — not another metaphor. The cancer-as-invasion metaphor has dominated oncology for decades and has misdirected therapy toward eradicating moving fronts in diseases whose underlying ecology is bounded, internally regulated, and amenable to steering. The cancer-as-ecosystem metaphor, if it remains a metaphor rather than becoming a computable model, will repeat the same mistake at a higher level of rhetorical sophistication. The Cancer Ecology Center program is built on the conviction that the field now has the tools to move from metaphor to model — and that the scientific and clinical obligation is to do so.

10.4 The Center Is Built to Deliver

The Cancer Ecology Center at Johns Hopkins exists to make this program computable, testable, and clinically actionable. It has the intellectual architecture (the three-layer program and its mathematical connections), the instruments (ExposoGraph, GeoToxGraph, IsoMCED, CEDT, Prairie-Dogs simulator), the cohort platform (BGES, All of Us adapter, MIO Consortium), the institutional base (the Brady Urological Institute, 16-institution Cancer Ecology Consortium), and the commercial translation path (Kreftect, Inc. for IsoMCED). What it requires next is prospective validation — the sustained clinical and scientific effort that tests the three falsifiable predictions against real patient data.

The compact is simple: steer rather than eradicate, at every scale, with tools that are computable, falsifiable, and open to the world.

References

Note: The following is a representative citation pool drawn from the source manuscripts. A complete Vancouver-style numbered reference list will be compiled from CEDT v6.5, ExposoGraph v12, Prairie-Dog Perspective, Evolutionary Contrast Mapping, and CarcinoGenomics R01 references_v7 as the source pool, targeting 120-150 references.

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Appendix A: Formalism Glossary

Bayesian Data Assimilation. A framework for sequentially updating a probability distribution (the posterior) over a latent state by combining a prior prediction from a dynamical model with the likelihood of new observations. In the CEDT, the latent state $X(t)$ is the quantity being estimated; the dynamical model (GLV) propagates the prior between observations; and an Extended, Ensemble, or Unscented Kalman Filter updates the posterior at each observation time. The result is a continuously maintained probability distribution over the tumor ecological state, not a point estimate.

EKF / EnKF / UKF (Extended, Ensemble, Unscented Kalman Filter). Three implementations of Bayesian data assimilation for nonlinear dynamical systems. The EKF linearizes the nonlinear model at each assimilation step (fast but potentially inaccurate for strongly nonlinear systems). The EnKF propagates an ensemble of state realizations and avoids linearization (computationally heavier but more accurate for nonlinear systems). The UKF uses deterministically chosen sigma points to capture higher-order moments of the nonlinear distribution

(a balance between EKF speed and EnKF accuracy). The CEDT implements all three; choice among them is a parameter of deployment.

G-Function (Fitness-Generating Function). $G(v; u, x, E)$ is the per-capita growth rate of an individual playing strategy v in a population with resident strategies u and abundances x , embedded in environmental state E . Defined by Vincent and Brown [Vincent & Brown 2005]. Two coupled equations follow: an ecological dynamic $dx_i/dt = x_i \cdot G(v; u, x, E) | \{v=u_i\}$, and an evolutionary dynamic du_i/dt proportional to $\partial G/\partial v | \{v=u_i\}$. Strategies at evolutionarily stable strategy (ESS) maxima persist. The G-function separates strategy (cell phenotype) from circumstance (microenvironment) and grounds adaptive therapy as a control problem.

Generalized Lotka-Volterra (GLV). A system of ordinary differential equations governing the dynamics of n interacting populations: $dN_i/dt = r_i(\text{state}) \cdot N_i \cdot [1 - \sum_j (A_{ij} \cdot N_j) / K_i]$. In the CEDT, population abundances N_i include tumor clones, anti-tumor immune cells, pro-tumor immune cells, stromal cells, and vascular compartments. The interaction matrix $A(t)$ captures who suppresses whom and at what rates; it is treated as latent and re-estimated by Bayesian data assimilation at each observation time. Resource coupling adds explicit oxygen (V) and habitat quality (Q_{hab}) compartments that gate growth rates through state-dependent functions.

Interaction Matrix $A(t)$. In a GLV system of n populations, A is an $n \times n$ matrix where entry A_{ij} is the per-capita effect of population j on the growth rate of population i . Positive A_{ij} (competition): j suppresses i . Negative A_{ij} (facilitation): j promotes i . In the CEDT, A is not fixed; it is treated as a latent quantity updated at each clinical visit through Bayesian data assimilation. The time-variation captures immune phenotype shifts, drug-induced selection, and ecological regime transitions.

Michaelis-Menten Flux Engine. In ExposoGraph 2.0, carcinogen metabolic flux through each enzyme is computed using Michaelis-Menten kinetics: $\text{flux} = (V_{\text{max}} \cdot [\text{substrate}]) / (K_m + [\text{substrate}])$, with V_{max} and K_m modified by genotype-dependent activity variants for CYP, NAT, GST, ALDH, SULT, and UGT isoforms. Competitive inhibition between co-metabolized carcinogens is modeled by the appropriate denominator extension. The resulting flux ratios span three orders of magnitude across genotype combinations.

Sobol Sensitivity Analysis. A variance-based global sensitivity method that decomposes the output variance of a model into contributions from each input parameter and their interactions. First-order indices S_i quantify the contribution of parameter i alone; total-effect indices ST_i quantify its contribution including all interactions with other parameters. In the CEDT, Sobol analysis over the eight non-assimilated fixed parameters confirms that absolute-abundance scaling (x_i cellularity) dominates forecast uncertainty, with first-order $S_i = 0.95$ for time-to-progression and 1.00 for 12-month log-PSA.

State-Space Framework. A mathematical representation of a dynamical system in which the system's behavior is captured by a latent state vector $X(t)$ that evolves according to a specified dynamical model (the state equation), and is related to observable quantities by an observation model (the measurement equation). In the CEDT, the state equation is the GLV system; the measurement equation maps the latent state to clinical observations (imaging volume, ctDNA VAF, tissue genomics, laboratory panels, TME readout). The state-space framework is the natural setting for Bayesian data assimilation.

Appendix B: Program Asset Map

Layer	Asset	URL / DOI	Status
Exposure	ExposoGraph 2.0 manuscript + code	github.com/kpienta-cloud/Exposograph	Under review, PLOS Computational Biology
Contrast	GeoToxGraph + evolutionary contrast mapping manuscript	github.com/kpienta-cloud/geobacter ; cloud.github.io/geobacter/	Manuscript in revision; browser live
Behavior	Prairie-Dogs G-function simulator + Perspective manuscript	github.com/kpienta-cloud/Prairie-Dogs ; cloud.github.io/Prairie-Dogs/	Perspective in revision; simulator live
Forecast			v6.5 manuscript;

Layer	Asset	URL / DOI	Status
Detection	CEDT framework + manuscript (v6.5)	github.com/kpienta-cloud/CEDT	submission imminent
	IsoMCED assay + Krefect, Inc.	Biswas Fast Grant funded; CAP/CLIA prep underway	Pilot funded; commercial translation path active
Origin / R01	CarcinoGenomics R01 (Research Strategy v61)	NIH submission	v61 Research Strategy complete; submission active
Center	Cancer Ecology Center + 16-institution Consortium	cancerecology.com	Consortium operational
Population cohort	Hopkins BGES (Biobank for Genomics and Exposomics)	R01 in progress for expansion to n = 1,000	Enrolling; R01 submission active
Validation cohort	All of Us adapter (ExposoGraph)	allofus.nih.gov integration	Adapter built; pre-registration in progress
International	MIO Consortium (Lund, Aristotle, Copenhagen, Hopkins)	Consortium agreement active	Multi-site collaboration active

All code repositories are or will be released under MIT license; all datasets under CC-BY 4.0; all preprints submitted to bioRxiv or medRxiv simultaneously with journal submission. Zenodo DOIs will be minted for each v1.0 release of each component.